```java
import java.io.*;
import java.util.ArrayList;

public class GradeAnalyzer {

  public static void main(String[] args) {
    // Step 1: read scores from file
    ArrayList<Integer> scores = readScores("scores.txt");

    // Handle empty file
    if (scores.isEmpty()) {
      System.out.println("No valid scores found in file.");
      return;
    }

    // Step 2: calculate statistics
    double average = calculateAverage(scores);

    // Find highest and lowest scores
    int highest = Integer.MIN_VALUE;
    int lowest = Integer.MAX_VALUE;

    for (int score : scores) {
      if (score > highest) {
        highest = score;
      }
```

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    if (score < lowest) {
        lowest = score;
    }
}

// Count grade bands
int countA = 0; // 90+
int countB = 0; // 80-89
int countC = 0; // 70-79
int countD = 0; // 60-69
int countF = 0; // below 60

for (int score : scores) {
    if (score >= 90) {
        countA++;
    } else if (score >= 80) {
        countB++;
    } else if (score >= 70) {
        countC++;
    } else if (score >= 60) {
        countD++;
    } else {
        countF++;
    }
}
```

```
    // Step 3: write and print report
    writeReport(scores, average, highest, lowest, "report.txt",
            countA, countB, countC, countD, countF);
}

// Returns a list of valid scores read from the file
public static ArrayList<Integer> readScores(String filename) {
    ArrayList<Integer> scores = new ArrayList<>();
    int invalidLineCount = 0;

    try (BufferedReader reader = new BufferedReader(new FileReader(filename))) {
        String line;
        while ((line = reader.readLine()) != null) {
            line = line.trim();

            // Skip blank lines
            if (line.isEmpty()) {
                continue;
            }

            // Try to parse the line as an integer
            try {
                int score = Integer.parseInt(line);
                scores.add(score);
            } catch (NumberFormatException e) {
                System.out.println("Warning: Invalid line '" + line + "' skipped");
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                invalidLineCount++;
            }
        }
    } catch (IOException e) {
        System.out.println("Error reading file: " + e.getMessage());
    }

    if (invalidLineCount > 0) {
        System.out.println("Total invalid lines skipped: " + invalidLineCount);
    }

    return scores;
}

// Returns the average of a list of scores, or 0.0 if the list is empty
public static double calculateAverage(ArrayList<Integer> scores) {
    if (scores.isEmpty()) {
        return 0.0;
    }

    double sum = 0.0;
    for (int score : scores) {
        sum += score;
    }

    return sum / scores.size();
```

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}

// Writes and prints the report
public static void writeReport(ArrayList<Integer> scores,
                               double avg, int high, int low,
                               String outputFile,
                               int countA, int countB, int countC,
                               int countD, int countF) {

    try (BufferedWriter writer = new BufferedWriter(new FileWriter(outputFile))) {
        // Format the report
        String line1 = String.format("=== Grade Analysis Report ===%n");
        String line2 = String.format("Total scores processed: %d%n", scores.size());
        String line3 = String.format("%n");
        String line4 = String.format("Average score:   %.2f%n", avg);
        String line5 = String.format("Highest score:   %d%n", high);
        String line6 = String.format("Lowest score:    %d%n", low);
        String line7 = String.format("%n");
        String line8 = String.format("Grade distribution:%n");
        String line9 = String.format("  A (90-100):   %d%n", countA);
        String line10 = String.format("  B (80-89):    %d%n", countB);
        String line11 = String.format("  C (70-79):    %d%n", countC);
        String line12 = String.format("  D (60-69):    %d%n", countD);
        String line13 = String.format("  F (below 60): %d%n", countF);

        // Write to file
```

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writer.write(line1);
writer.write(line2);
writer.write(line3);
writer.write(line4);
writer.write(line5);
writer.write(line6);
writer.write(line7);
writer.write(line8);
writer.write(line9);
writer.write(line10);
writer.write(line11);
writer.write(line12);
writer.write(line13);

// Print to terminal
System.out.println("\n" + line1);
System.out.println(line2);
System.out.print(line3);
System.out.print(line4);
System.out.print(line5);
System.out.print(line6);
System.out.print(line7);
System.out.print(line8);
System.out.print(line9);
System.out.print(line10);
System.out.print(line11);
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            System.out.print(line12);
            System.out.print(line13);


            System.out.println("Report written to " + outputFile);


        } catch (IOException e) {
            System.out.println("Error writing report: " + e.getMessage());
        }
    }
}
```